Malware Deep Analysis

Carson Blin

March 2026

1 Reviewed Literature

- Bo Yu et al., "A Survey of Malware Behavior Description and Analysis," *Frontiers of Information Technology Electronic Engineering* 19, no. 5 (2018): 583–603, https://doi.org/10.1631/FITEE.1601745
- Kshatriya Vinaya Sree Bai and M. Thirumaran, "Survey of Deep Learning Techniques for Malware Detection: Insights, Challenges, and Future Directions," in *2024 4th International Conference on Soft Computing for Security Applications (ICSCSA)* (2024), 320–324, https://doi.org/10.1109/ICSCSA64454.2024.00057
- Bhavesh Gilhare, Vikas Sihag, and Gaurav Dave, "A Brief Survey of Persistence Approaches in Real World Malwares," in *2025 13th International Conference on Intelligent Systems and Embedded Design (ISED)* (2025), 819–824, https://doi.org/10.1109/ISED67359.2025.11405257

2 Summary of Learning

Throughout these research papers, I learned more information specifically about malware protocols and the types of behaviors observed in malware. Additionally, I learned specifically about malware persistence and how the malware targets specific information in the OS such as modifying system files, creating malicious tasks, and modifying system registry files.

3 Project Development Progress

The overall development process of this project is nearing the Collection portion of the project and will be progressing into the Analysis stage. The environment that is being utilized throughout this development is Any.run as it provides a safe virtual environment to run this malware. I am also utilizing the MITRE ATTACK website, which is a knowledge base site regarding malware techniques and tactics that have been identified in past cyber attacks. Results from this experiment will need to be further analyzed and refined in the next development phase of this project.

4 Next-Phase Plan

In my next report, my next objectives are to perform more analysis on Any.run and look at more specific processes that malware is associated with. Further experiments will need to be conducted to ensure enough data is collected and accurate. and I also plan on refining the data observed in Any.run and create visuals like graphs from the data collected.